

CSCI3230 (ESTR3108)

Fundamentals of Artificial Intelligence

Lecture 5. Clustering Algorithms

Prof. Qi Dou

Email: qidou@cuhk.edu.hk

Office: Room 911, 9/F, SHB

Dept. of Computer Science & Engineering
The Chinese University of Hong Kong



Outline

Part 1. Clustering basics

Part 2. K-Means

Part 3. Density-based clustering

Part 4. Agglomerative hierarchical clustering

Part 5. Summary



Part 1. Clustering basics

Clustering basics

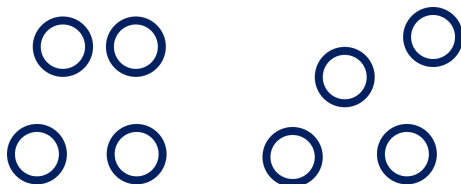
What is “Samples”?

- Samples represent a set of individuals or objects collected or selected from a statistical population by a defined procedure.

Clustering basics

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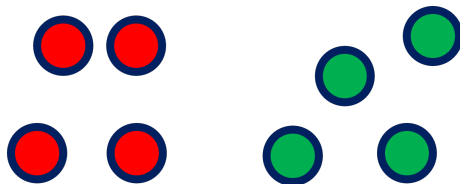
- Samples represent a set of individuals or objects collected or selected from a statistical population by a defined procedure.
- Each sample can be called a **Data Point**.



Clustering basics

What is “Labels”?

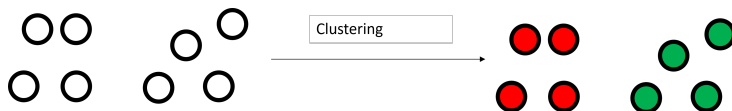
- A category or class applied to an object or a thing.
- For example, there are two classes (indicated by two colors)
 - red for Class 0 and green for Class 1.



Clustering basics

What is Clustering?

- Group a set of objects so that objects in the same group (called a cluster) are more similar (in some sense) to each other than to those in other groups (clusters)¹.
- High intra-cluster similarity and low inter-cluster similarity.
- Clustering belongs to **unsupervised learning** (i.e., having samples but without using labels).

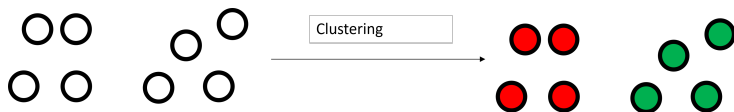


¹https://en.wikipedia.org/wiki/Cluster_analysis

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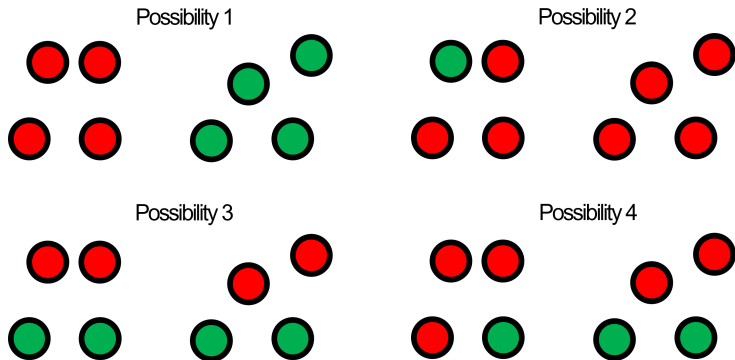


- In comparison, classification belongs to supervised learning (i.e., need both samples and corresponding labels).

¹https://en.wikipedia.org/wiki/Cluster_analysis

Example: 2D data points

- Just show four possible clustering results in the following:



Measure of similarity - Euclidean distance (L_2 -norm)

How to measure whether two data points are “similar”?

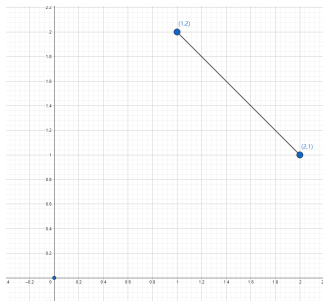
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- **Euclidean distance:** $\text{dist}(\mathbf{a}, \mathbf{b}) = \sqrt{\sum_{i=1}^n (a_i - b_i)^2}$
- For example, in 2D space, when $\mathbf{a} = (1, 2)$ and $\mathbf{b} = (2, 1)$,
 $\text{dist}(\mathbf{a}, \mathbf{b}) = \sqrt{(a_1 - b_1)^2 + (a_2 - b_2)^2} = \sqrt{(1 - 2)^2 + (2 - 1)^2} = \sqrt{2}$

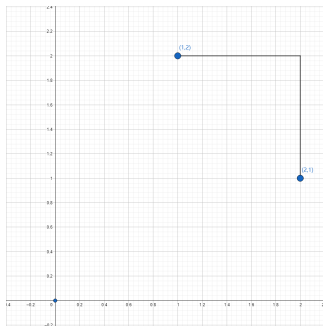


Measure of similarity - Manhattan distance (L_1 -norm)

How to measure whether two data points are “similar”?

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 $\text{dist}(\mathbf{a}, \mathbf{b}) = |a_1 - b_1| + |a_2 - b_2| = |1 - 2| + |2 - 1| = 2$



Measure of similarity - Chebyshev distance

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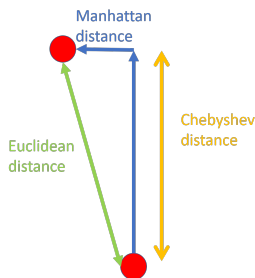
- **Chebyshev distance:** $\text{dist}(\mathbf{a}, \mathbf{b}) = \max_i (|a_i - b_i|)$
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Measure of similarity - Minkowski distance (p-norm)

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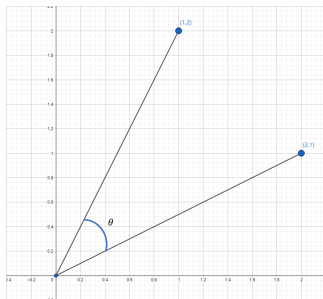
- **Minkowski distance:** $\text{dist}(\mathbf{a}, \mathbf{b}) = (\sum_{i=1}^n |a_i - b_i|^p)^{\frac{1}{p}}$
- when p is 1, Minkowski distance $\rightarrow$ Manhattan distance
- when p is 2, Minkowski distance $\rightarrow$ Euclidean distance
- when p is infinite, Minkowski distance $\rightarrow$ Chebyshev distance

Measure of similarity - Cosine similarity

How to measure whether two data points are “similar”?

Given two data points $\mathbf{a} = (a_1, a_2, \dots, a_n)$, $\mathbf{b} = (b_1, b_2, \dots, b_n) \in \mathbb{R}^n$.

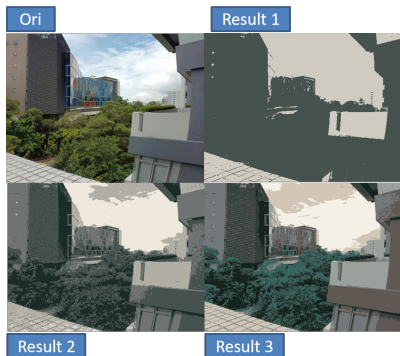
- **Cosine similarity:** $\text{dist}(\mathbf{a}, \mathbf{b}) = \cos(\theta) = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\| \cdot \|\mathbf{b}\|} = \frac{\sum_{i=1}^n (a_i \cdot b_i)}{\sqrt{\sum_{i=1}^n a_i^2} \sqrt{\sum_{i=1}^n b_i^2}}$
- For example, in 2D space, when $\mathbf{a} = (1, 2)$ and $\mathbf{b} = (2, 1)$,
 $\text{dist}(\mathbf{a}, \mathbf{b}) = \frac{1 \times 2 + 2 \times 1}{\sqrt{1^2 + 2^2} \times \sqrt{2^2 + 1^2}} = \frac{4}{5} = 0.8$



Example of clustering applications

Image processing (e.g., semantic segmentation)

- different results from different settings of clustering hyper-parameters.



Clustering algorithms

- Partitional clustering algorithms
 - ① Centroid-based: a cluster is characterized and represented by a central vector, and data points that are close to these vectors are assigned to the respective clusters. (e.g., **K-Means**)

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- ④ Spectral Clustering

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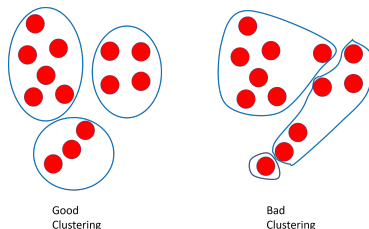
- Hierarchical clustering algorithms

- ① **Agglomerative** (bottom up)
- ② Divisive (top-down)

- ...

Cluster evaluation (intuitively)

- Intra-cluster cohesion (compactness)
 - Cohesion measures how near the data points in a cluster are to the cluster centroid.
- Inter-cluster separation (isolation)
 - Separation means that different cluster centroids should be far away from one another.



- In most applications, expert judgements are still the key.



Part 2. K-Means

K-Means

- K-Means (MacQueen, 1967) is a partitional clustering algorithm.
- Let the set of data points be $\{\mathbf{a}^{(1)}, \mathbf{a}^{(2)}, \dots, \mathbf{a}^{(M)}\}$, where $\mathbf{a}^{(m)} = (a_1^{(m)}, a_2^{(m)}, \dots, a_n^{(m)})$ is a n -dimensional vector in $\mathbf{a} \in \mathbb{R}^n$
- The K-Means algorithm partitions the given data into K clusters:
 - Each cluster has a cluster center (a.k.a. centroid)
 - K is manually specified by the user

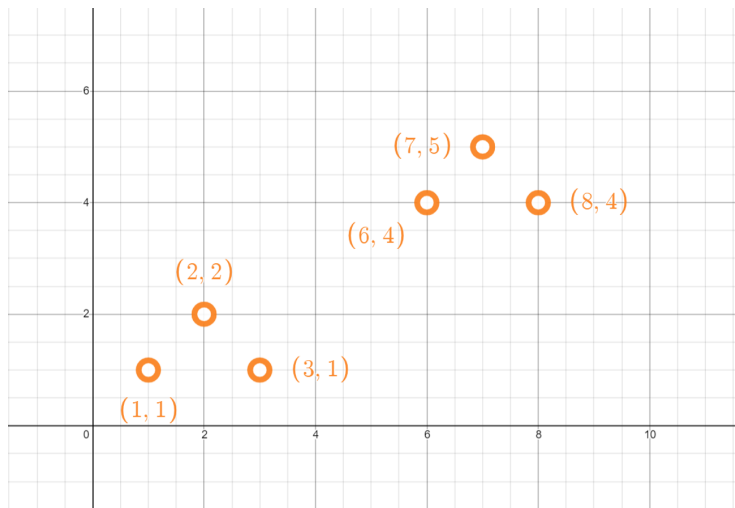
K-Means

Intuitively, the K-Means algorithm works as follows:

- 1 Choose K (random) data points (as seeds) to be the initial **centroids** as cluster centers.
- 2 Assign each data point to the closest centroid.
- 3 Re-compute the centroids using the current cluster memberships.
- 4 If a convergence criterion is not met, repeat steps 2 and 3.

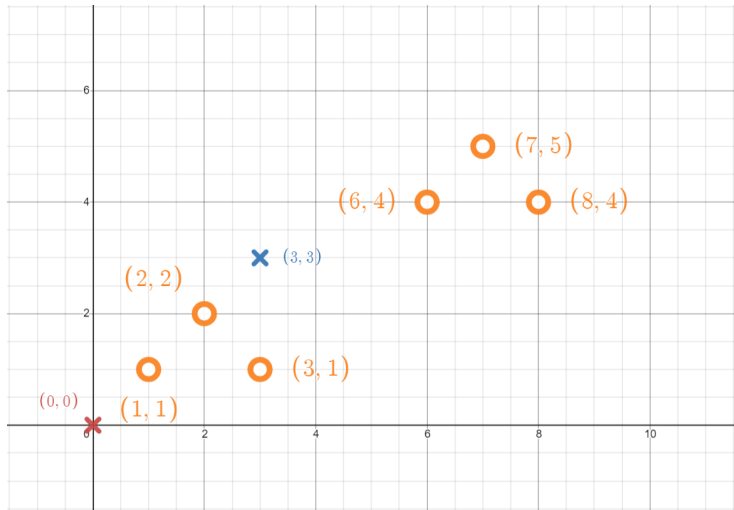
K-Means - an example

K-Means is an iterative clustering algorithm.



K-Means

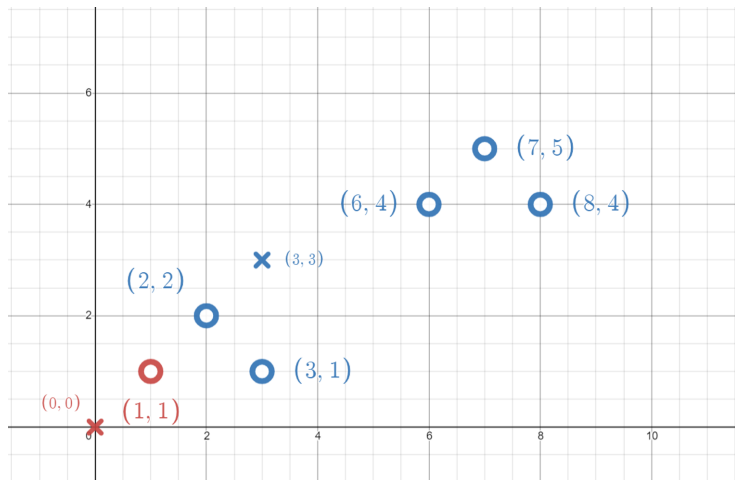
Initialize: Pick K (number of clusters) random points as cluster centroids. Assume $K=2$ and random centroids are $(0,0)$ and $(3,3)$



K-Means

Iteration: Assign points to the closest cluster centroid (Euclidean Dist).

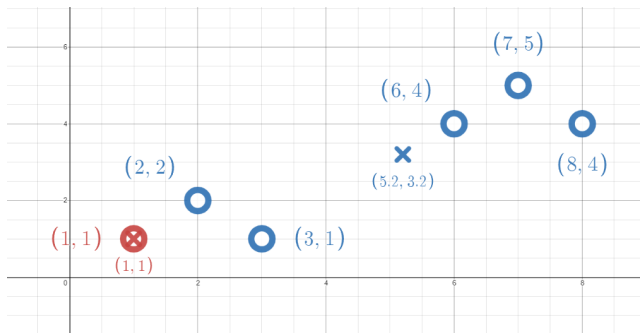
- Cluster 0: (1,1)
- Cluster 1: (2,2), (3,1), (6,4), (7,5), (8,4)



K-Means

Iteration: Update the cluster centroid to the mean of its assigned points.

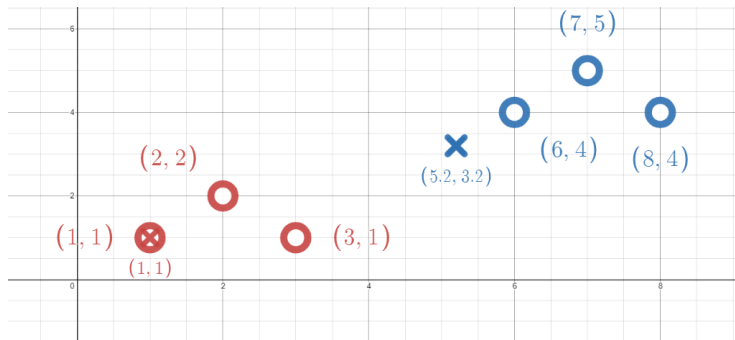
- Cluster 0: $(1,1) \rightarrow$ cluster 0 centroid is $(1,1)$
- Cluster 1: $(2,2), (3,1), (6,4), (7,5), (8,4) \rightarrow$ cluster 1 centroid is $((2 + 3 + 6 + 7 + 8)/5, (2 + 1 + 4 + 5 + 4)/5) = (5.2, 3.2)$



K-Means

Iteration: Assign data points to the (new) closest cluster centroids.

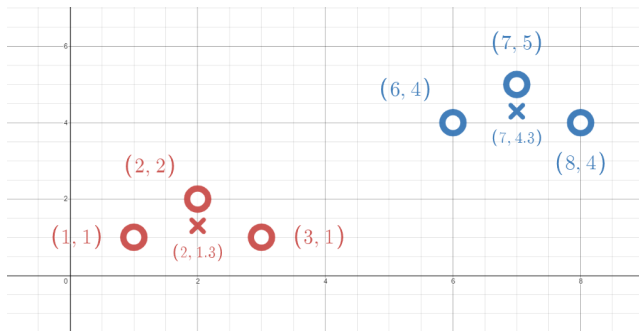
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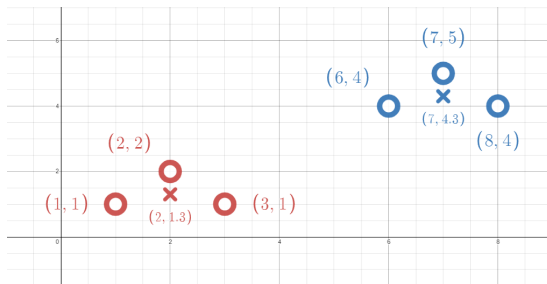
- Cluster 0: $(1,1), (2,2), (3,1) \rightarrow$ cluster 0 centroid is $((1 + 2 + 3)/3, (1 + 2 + 1)/3) = (2, 1.3)$
- Cluster 1: $(6,4), (7,5), (8,4) \rightarrow$ cluster 1 centroid is $((6 + 7 + 8)/3, (4 + 5 + 4)/3) = (7, 4.3)$



K-Means

Iteration: Assign data points to the (new) closest cluster centroid.

- Cluster 0: $(1,1)$, $(2,2)$, $(3,1)$.
- Cluster 1: $(6,4)$, $(7,5)$, $(8,4)$.
- No change anymore, therefore, the clustering is **finished**.



K-Means convergence (stopping) criteria

- no (or minimum) reassignment of data points to different clusters, or
- no (or minimum) change of centroids, or
- minimum decrease in the sum of squared error (SSE)
 - C_i is the i -th cluster
 - μ_i is centroid of cluster C_i (the mean vector of all data points in C_i)
 - $\text{dist}(\mathbf{x}, \mu_i)$ is the Euclidean distance between data point $\mathbf{x}$ and centroid μ_i :

$$\text{SSE} = \sum_{i=1}^K \sum_{\mathbf{x} \in C_i} \text{dist}(\mathbf{x}, \mu_i)^2$$

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$$\text{SSE} = \sum_{i=1}^K \sum_{x \in C_i} \text{dist}(x, \mu_i)^2$$

K-Means algorithm:

- 1 Select K points as the initial centroids.
- 2 **repeat**
 - 1) Form K clusters by assigning all points to the closest centroid.
 - 2) Recompute the centroid of each cluster
- 3 **until** fulfill the stopping condition

K-Means complexity analysis

- M = number of data points, K = number of clusters, I = number of iterations, N = number of attributes (feature dimension).
- Guaranteed to converge in a finite number of iterations.
- Time complexity:
 - 1 Running time per iteration:
 - assign data points to closest cluster center: $O(KM)$
 - change the cluster center to the average of its assigned points: $O(M)$
 - 2 For I iterations: $O(IKM)$
 - 3 For N -dimensional vectors: $O(IKMN)$

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 - 2 For I iterations: $O(IKM)$
 - 3 For N -dimensional vectors: $O(IKMN)$
- Space complexity:
 - 1 Store N points: $O(MN)$
 - 2 Store K centroids: $O(KN)$
 - 3 Overall space: $O((M+K)N)$

K-Means advantages

K-means is the most popular clustering algorithm

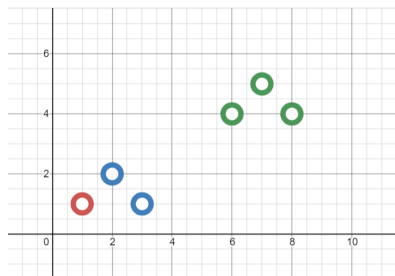
- Easy to understand and to implement
- Efficient, if K and I are small, considered as a linear algorithm
- Scale to large data sets
- Guarantee convergence
- Clusters can be easily interpreted and even visualized
- May produce tighter clusters than hierarchical clustering

K-Means disadvantages

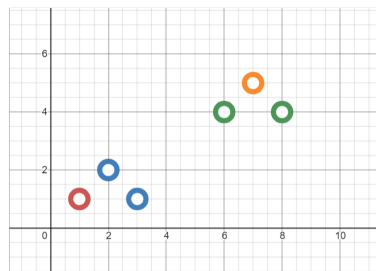
K-Means needs to manually set hyper-parameter K.

- Sensitive to the hyper-parameter of pre-assigned number of clusters.
- Different K would lead to different clustering results.

K=3



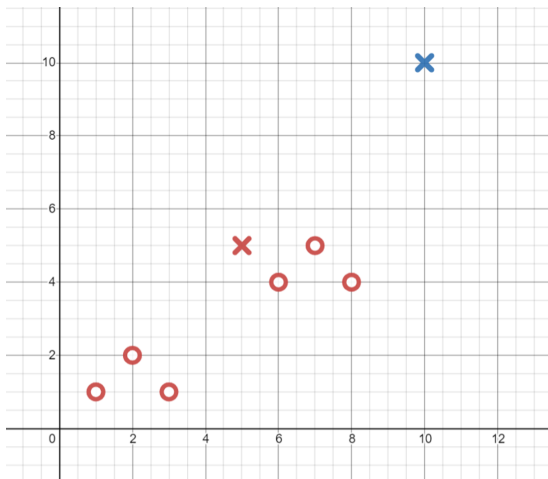
K=4



K-Means disadvantages

K-Means may encounter empty cluster due to improper initialization.

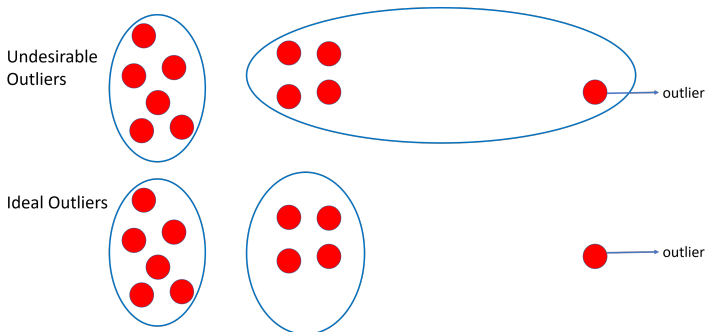
- Assume initial centroids are (5,5) and (10,10)
- All data points will be assigned to cluster with the centroid (5,5)



K-Means disadvantages

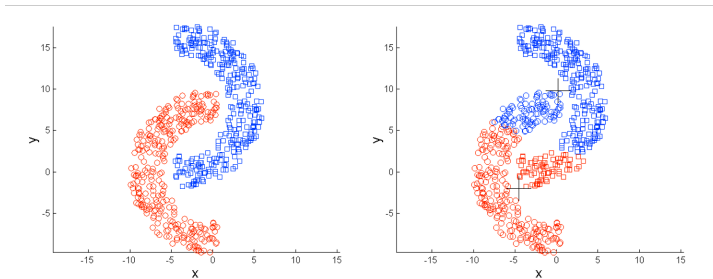
K-Means algorithm is sensitive to outliers.

- Outliers are data points that are very far away from other data points.
- Outliers could be errors in the data recording or some special data points with very different values.



K-Means disadvantages

- K-Means may not work well for data points with non-globular shapes.
- Because it defines clusters based on Euclidean distance to centroid.



Original Points

K-means (2 Clusters)

K-Means quiz

- Suppose we have three cluster centroids $\mu_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $\mu_2 = \begin{bmatrix} -3 \\ 0 \end{bmatrix}$ and $\mu_3 = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$. Furthermore, we have a training example $x^{(i)} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$.

After a cluster assignment step, which cluster will $x^{(i)}$ be assigned?

- 1 Cluster 1
- 2 Cluster 2
- 3 Cluster 3
- 4 I Do Not Know

K-Means quiz

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- 1 Cluster 1
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Answer: A

Reason: The distance between μ_1 and $x^{(i)}$ is the smallest, compared to μ_2 and μ_3



Part 3. Density-based clustering

Density-based clustering

- Clusters are dense regions in the data space
- A cluster is defined as a maximal set of density-connected points
- Noise points lie in regions of low density
- Discover clusters of arbitrary shape and size

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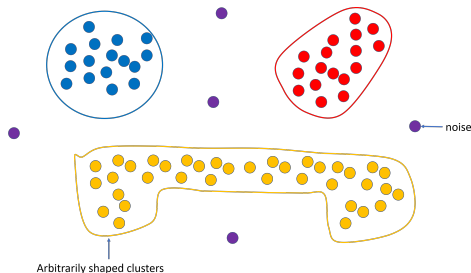
DBSCAN



k-means



DBSCAN: Density-based spatial clustering of applications with noise



[PDF] [A density-based algorithm for discovering clusters in large spatial databases with noise.](#)

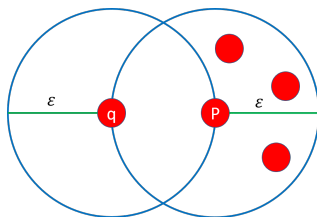
[M Ester](#), [HP Kriegel](#), [J Sander](#), [X Xu](#) - kdd, 1996 - aaai.org

Clustering algorithms are attractive for the task of class identification in spatial databases. However, the application to large spatial databases rises the following requirements for clustering algorithms: minimal requirements of domain knowledge to determine the input parameters, discovery of clusters with arbitrary shape and good efficiency on large databases. The well-known clustering algorithms offer no solution to the combination of these requirements. In this paper, we present the new clustering algorithm DBSCAN relying ...

☆ 被引用次数: 22071 相关文章 所有 82 个版本

DBSCAN: Density-based spatial clustering of applications with noise

- Two parameters:
 - ϵ : maximum radius of the neighborhood
 ϵ -Neighbor: data points within a radius of ϵ from a data point (including the point itself)
 - *MinPts*: minimum number of points required in an ϵ -Neighbor
- Density definition:
 - density = number of points within a specified radius ϵ
 - “high density”: data point's ϵ -Neighbor contains at least *MinPts* data



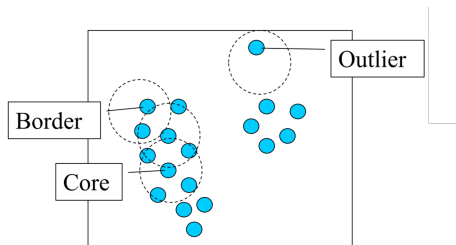
MinPts = 4

ϵ – Neighborhood of p
 ϵ – Neighborhood of q
Density of p is “high” (*MinPts* = 4)
Density of q is “low” (*MinPts* = 4)

DBSCAN

Given ϵ and $MinPts$, categorize the data points to three exclusive groups.

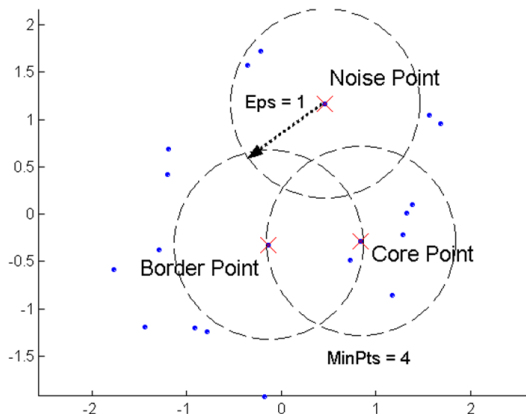
- **Core point:** has more than or equal to $MinPts$ within ϵ . These are points that are at the interior of a cluster.
- **Border point:** has fewer than $MinPts$ within ϵ , but is in the neighborhood of a core point.
- **Noise point (or outlier):** any point that is neither a core point nor a border point.



$\epsilon = 1\text{unit}$, $MinPts = 5$

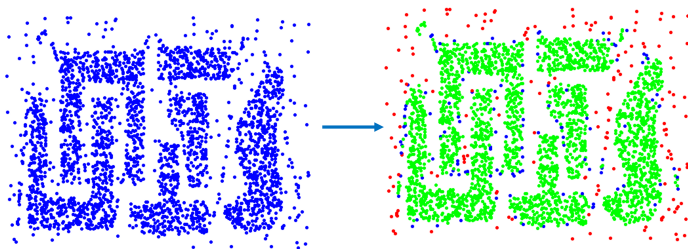
DBSCAN

Example of using ϵ and $MinPts$ to categorize data points.



DBSCAN

Example of using ϵ and $MinPts$ to categorize data points.



Original Points

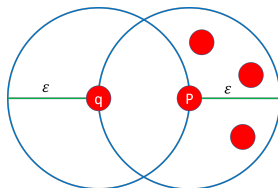
Point types: **core**, **border**
and **noise**

Density-reachable

Directly density-reachable

- An object q is directly density-reachable from object p if p is a core object and q is in p 's ϵ -neighborhood.

In the following example:



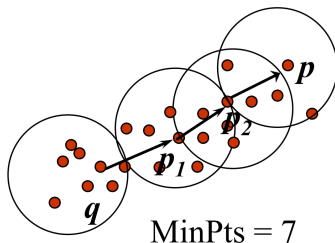
$MinPts = 4$

- q is directly density-reachable from p
- p is not directly density-reachable from q
- Note that density-reachability is asymmetric

Density-connected

Indirectly density-reachable (a.k.a. density-connected)

- A point p is directly density-reachable from p_2
- p_2 is directly density-reachable from p_1
- p_1 is directly density-reachable from q
- $p \leftarrow p_2 \leftarrow p_1 \leftarrow q$ form a chain
- p is (indirectly) density-reachable from q
(a.k.a. p is density-connected from q)
- q is not density-reachable from p (why?)



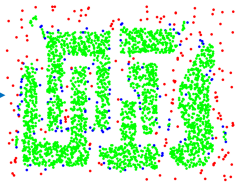
DBSCAN algorithm

- Label data points into core, border and noise
- Eliminate noise points
- For every core point p that has not been assigned to a cluster
 - Create a new cluster with the point p and all the points that are density-connected to p
- Assign border points to the cluster of the closest core point.

DBSCAN results visualization



Original Points



Point types: core, border
and noise



Final Cluster

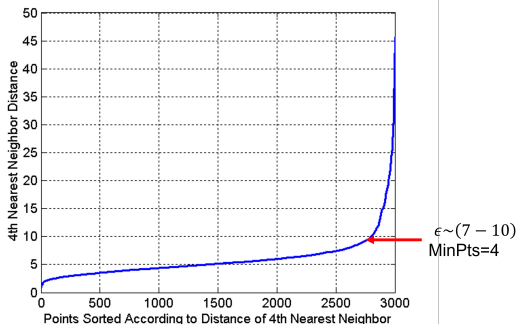
DBSCAN algorithm principles

DBSCAN relies on a density-based notion of cluster: A cluster is defined as a maximal set of density-connected points.

- Principle 1: Each cluster contains at least one core point.
- Principle 2: Given any two core points p and q , if $N(p)$ contains q (or $N(q)$ contains p), then p and q are in the same cluster.
- Principle 3: A border point p is assigned arbitrarily to one of the clusters containing these core points.
- Principle 4: All noise points do not belong to any clusters.

DBSCAN: determining ϵ and MinPts

- The idea is that, for points in a cluster, their k^{th} nearest neighbors are at roughly the same distance.
- Noise points have the k^{th} nearest neighbor at a farther distance.
- So, plot sorted distance of every point to its k^{th} neighbor.
- Find the distance d where there is a “Knee” in the curve.

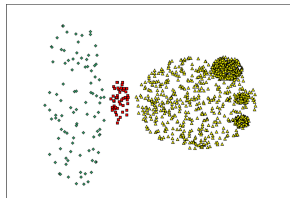
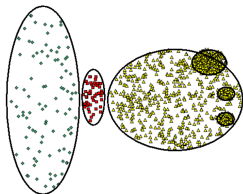


DBSCAN advantages

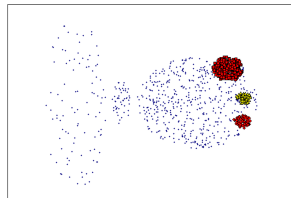
- DBSCAN can handle clusters of arbitrary shapes and sizes.
- Number of clusters is determined automatically. (c.f. K-Means)
- Can separate clusters from noise. Robust to outlier.
- Also a very commonly used clustering algorithm.

DBSCAN disadvantages

- Sensitive to parameters – hard to determine the optimal parameters.
- Has problem if identify clusters of varying densities.
- DBSCAN may not work well in high-dimensional data.
- Density estimation is kind of simplistic (i.e., doesn't create a real density function, but rather a graph of density-connected points).



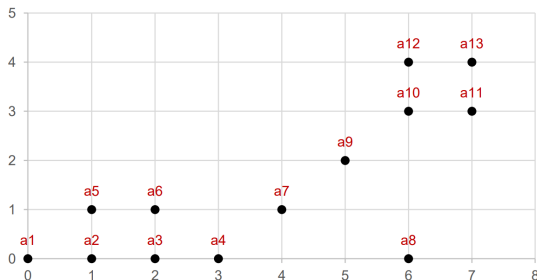
(MinPts=4, Eps=9.75).



(MinPts=4, Eps=9.92)

Quiz

Apply the DBSCAN clustering method to group the following data points:
 a_1 (0, 0), a_2 (1, 0), a_3 (2, 0), a_4 (3, 0), a_5 (1, 1), a_6 (2, 1), a_7 (4, 1),
 a_8 (6, 0), a_9 (5, 2), a_{10} (6, 3), a_{11} (7, 3), a_{12} (6, 4), a_{13} (7, 4)

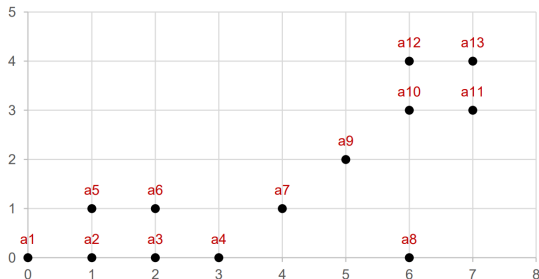


Quiz

Assume that parameters $\epsilon = \sqrt{2}$ and $\text{MinPts} = 4$ (use Euclidean distance)
Please answer the following questions.

1) What is the ϵ -neighbor of a_1 ?

- A. $\{a_1\}$
- B. $\{a_2, a_5\}$
- C. $\{a_2, a_5, a_6\}$
- D. $\{a_1, a_2, a_5\}$



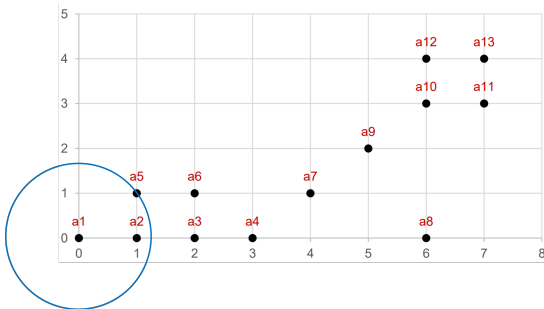
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- B. $\{a_2, a_5\}$
- C. $\{a_2, a_5, a_6\}$
- D. $\{a_1, a_2, a_5\}$

Correct Answer: D

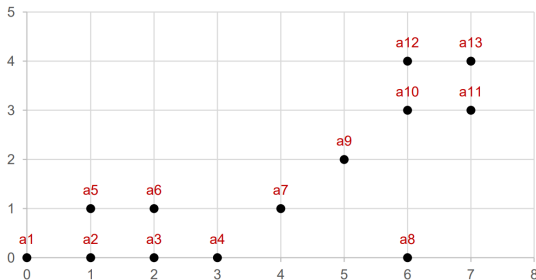


Quiz

Assume that parameters $\epsilon = \sqrt{2}$ and MinPts = 4 (use Euclidean distance)
Please answer the following question.

2) What are final cluster results?

- A. $\{a_1, a_2, a_3, a_4, a_5, a_6, a_7\}$, and $\{a_9, a_{10}, a_{11}, a_{12}, a_{13}\}$
- B. $\{a_1, a_2, a_3, a_4, a_5, a_6, a_7, a_8\}$, and $\{a_9, a_{10}, a_{11}, a_{12}, a_{13}\}$
- C. $\{a_1, a_2, a_3, a_4, a_5, a_6, a_7\}$, and $\{a_8, a_9, a_{10}, a_{11}, a_{12}, a_{13}\}$
- D. $\{a_1, a_2, a_3, a_4, a_5, a_6\}$, and $\{a_7, a_9, a_{10}, a_{11}, a_{12}, a_{13}\}$



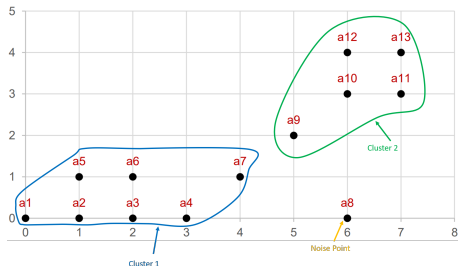
Quiz

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- B. $\{a_1, a_2, a_3, a_4, a_5, a_6, a_7, a_8\}$, and $\{a_9, a_{10}, a_{11}, a_{12}, a_{13}\}$
- C. $\{a_1, a_2, a_3, a_4, a_5, a_6, a_7\}$, and $\{a_8, a_9, a_{10}, a_{11}, a_{12}, a_{13}\}$
- D. $\{a_1, a_2, a_3, a_4, a_5, a_6\}$, and $\{a_7, a_9, a_{10}, a_{11}, a_{12}, a_{13}\}$

Correct Answer: A





Part 4. Agglomerative hierarchical clustering

Hierarchical clustering

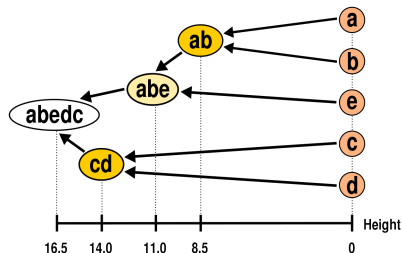
- Hierarchical clustering is a method of cluster analysis which seeks to build a hierarchy of clusters.

Hierarchical clustering

- Hierarchical clustering is a method of cluster analysis which seeks to build a hierarchy of clusters.
- Strategies for hierarchical clustering generally include two types:
 - Agglomerative: “bottom-up”, i.e., each observation starts in its own cluster, and pairs of clusters are merged as one moves up the hierarchy.
 - Divisive: “top-down”, i.e., all observations start in one cluster, and splits are performed recursively as one moves down the hierarchy.

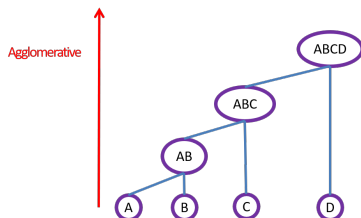
Hierarchical clustering

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 - Divisive: “top-down”, i.e., all observations start in one cluster, and splits are performed recursively as one moves down the hierarchy.
- The result of hierarchical clustering usually presented in dendrogram.



Agglomerative hierarchical clustering

- Agglomerative hierarchical clustering is a “bottom-up” approach
- Starting with each item in its own cluster, find the best pair to merge into a new cluster. Repeat until all clusters are fused together.

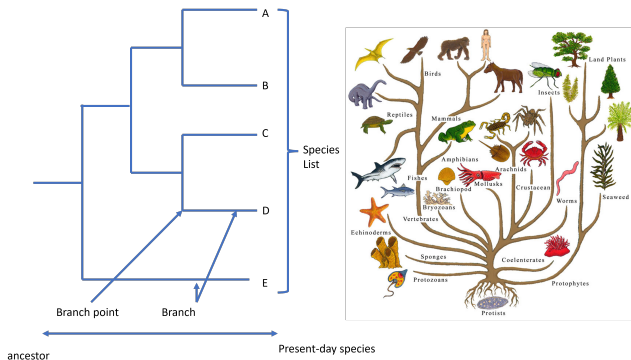


Basic algorithm is straightforward

1. Compute the distance matrix
2. Let each data point be a cluster
3. **Repeat**
4. **Merge the two closest clusters**
5. **Update the proximity matrix**
6. **Until** only a single cluster remains

Agglomerative hierarchical clustering - example

Phylogenetic tree in bioinformatics



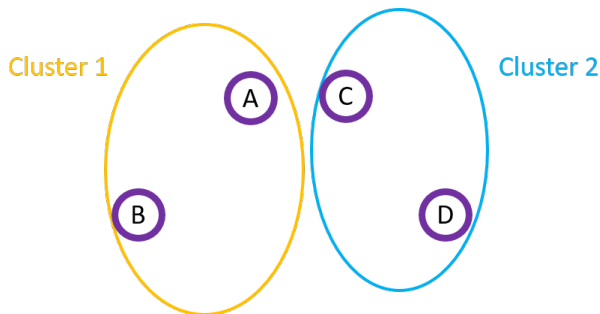
Distance matrix

- Create a distance matrix (a.k.a. proximity matrix) to record distance between clusters.
- A distance matrix is a square matrix (i.e., 2D array) containing the distances between every pair of elements of a set.
- The following is an example of a distance matrix for 5 data points, e.g., $\text{Dist}(\text{sample-2}, \text{sample-3}) = 5.0$ or $d_{2,3} = 5.0$

$$\begin{array}{c} \begin{matrix} & 1 & 2 & 3 & 4 & 5 \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{matrix} & \begin{pmatrix} 0.0 & & & & \\ 2.0 & 0.0 & & & \\ 6.0 & 5.0 & 0.0 & & \\ 10.0 & 9.0 & 4.0 & 0.0 & \\ 9.0 & 8.0 & 5.0 & 3.0 & 0.0 \end{pmatrix} \end{matrix} \end{array}$$

Distance matrix

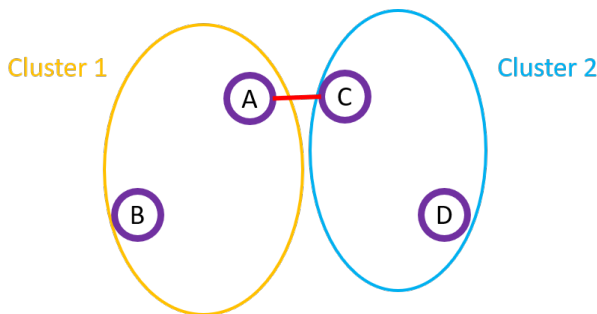
- Compute the distance between clusters (i.e., how to determine the best pair of clusters to merge?)
- For example: Cluster 1 has data points A and B; Cluster 2 has data points C and D.



Distance matrix

Min distance (a.k.a. Single Linkage)

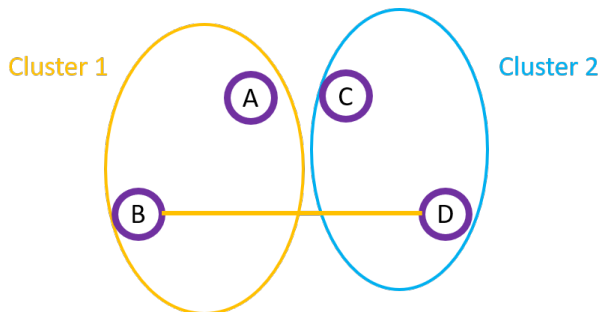
- Cluster distance = distance of two closest elements in each cluster
- $\text{dist}(\text{Cluster1}, \text{Cluster2}) = \min(\text{dist}(A, C), \text{dist}(A, D), \text{dist}(B, C), \text{dist}(B, D)) \implies \text{dist}(A, C)$



Distance matrix

Max distance (a.k.a. Complete Linkage)

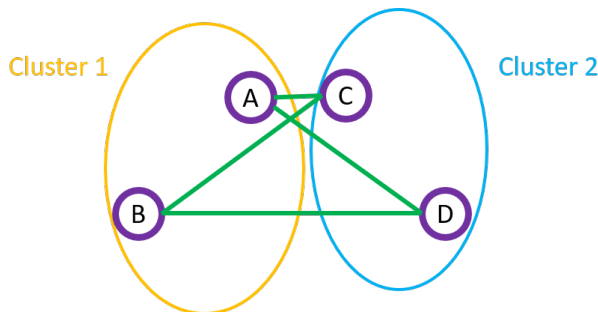
- Cluster distance = distance of two farthest elements
- $\text{dist}(\text{Cluster 1}, \text{Cluster 2}) = \max(\text{dist}(A,C), \text{dist}(A,D), \text{dist}(B,C), \text{dist}(B,D)) \implies \text{dist}(B,D)$



Distance matrix

Group average distance (a.k.a. Average Linkage)

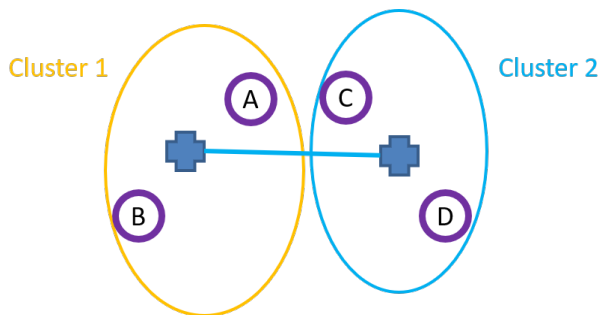
- Cluster distance = average distance of all pairs
- $\text{dist}(\text{Cluster 1}, \text{Cluster 2}) = (\text{dist}(A,C) + \text{dist}(A,D) + \text{dist}(B,C) + \text{dist}(B,D))/4$



Distance matrix

Distance between centroids (a.k.a. Centroid Linkage)

- Cluster distance = distance of two cluster centroids
- $\text{dist}(\text{Cluster 1}, \text{Cluster 2}) = \text{dist}(\frac{A+B}{2}, \frac{C+D}{2})$



A concrete example (min distance)

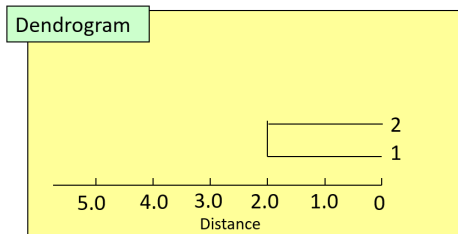
To cluster the data points with the afore-mentioned matrix distance (indicating min distance between two samples/clusters):

	1	2	3	4	5
1	0.0				
2	2.0	0.0			
3	6.0	5.0	0.0		
4	10.0	9.0	4.0	0.0	
5	9.0	8.0	5.0	3.0	0.0

A concrete example (min distance)

Step1: Merge the two closest clusters of $\{1\}$ and $\{2\}$ to one cluster $\{1,2\}$. Because based on the distance matrix, the distance between $\{1\}$ and $\{2\}$ is the smallest number.

	1	2	3	4	5
1	0.0				
2	2.0	0.0			
3	6.0	5.0	0.0		
4	10.0	9.0	4.0	0.0	
5	9.0	8.0	5.0	3.0	0.0

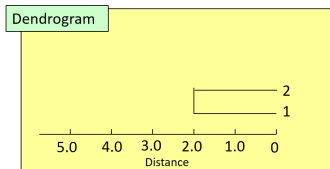
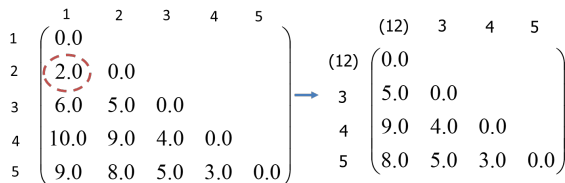


A concrete example (min distance)

Step2: Update the distance matrix. Form the new cluster $\{1,2\}$, and recalculate the min distance between all clusters.

- $d_{(1,2),3} = \min(d_{1,3}, d_{2,3}) = \min(6, 5) = 5$
- $d_{(1,2),4} = \min(d_{1,4}, d_{2,5}) = \min(10, 9) = 9$
- $d_{(1,2),5} = \min(d_{1,4}, d_{2,5}) = \min(9, 8) = 8$

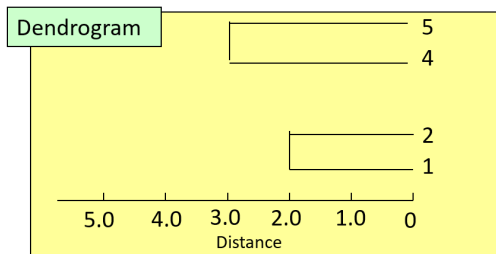
Hierarchical Clustering Methods-MIN Distance



A concrete example (min distance)

Step 3: Merge the two closest clusters of $\{4\}$ and $\{5\}$ to one cluster $\{4,5\}$. Because based on the distance matrix, the distance between $\{4\}$ and $\{5\}$ is the smallest number.

	(12)	3	4	5
(12)	0.0			
3	5.0	0.0		
4	9.0	4.0	0.0	
5	8.0	5.0	3.0	0.0

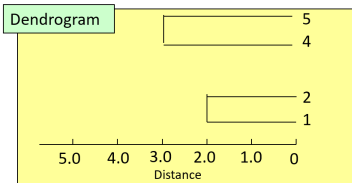


A concrete example (min distance)

Step 4: Update the distance matrix. Form the new cluster $\{4,5\}$, and recalculate the min distance between all clusters.

- $d_{(4,5),(1,2)} = \min(d_{4,(1,2)}, d_{5,(1,2)}) = \min(9, 8) = 8$
- $d_{(4,5),3} = \min(d_{4,3}, d_{5,3}) = \min(4, 5) = 4$

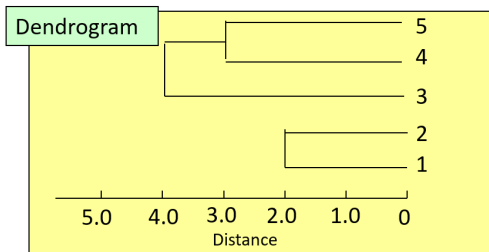
$$\begin{array}{c} (12) \quad 3 \quad 4 \quad 5 \\ \begin{pmatrix} (12) & 0.0 & & \\ 3 & 5.0 & 0.0 & \\ 4 & 9.0 & 4.0 & 0.0 \\ 5 & 8.0 & 5.0 & \textcolor{red}{3.0} & 0.0 \end{pmatrix} \end{array} \rightarrow \begin{array}{c} (12) \quad 3 \quad (4\ 5) \\ \begin{pmatrix} (12) & 0.0 & \\ 3 & 5.0 & 0.0 \\ (4\ 5) & 8.0 & 4.0 & 0.0 \end{pmatrix} \end{array}$$



A concrete example (min distance)

Step5: Merge the two closest clusters of $\{3\}$ and $\{4, 5\}$ to one cluster $\{3,4,5\}$. Because based on the distance matrix, the distance between 3 and 4, 5 is the smallest number.

$$\begin{array}{ccc} & (12) & 3 & (4\ 5) \\ (12) & \left(\begin{array}{ccc} 0.0 & & \\ 5.0 & 0.0 & \\ 8.0 & 4.0 & 0.0 \end{array} \right) & & \\ 3 & & & \\ (4\ 5) & & & \end{array}$$

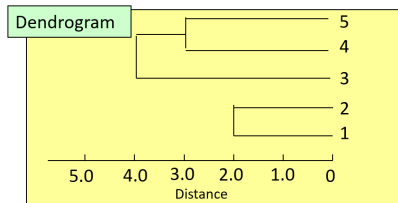


A concrete example (min distance)

Step 6: Update the distance matrix. Form the new cluster $\{3,4,5\}$, and recalculate the min distance between all clusters.

- $d_{((4,5),3),(1,2)} = \min(d_{(4,5),(1,2)}, d_{3,(1,2)}) = \min(8, 5) = 5$

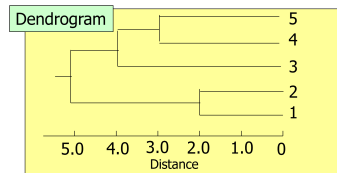
$$\begin{array}{c} (12) \quad 3 \quad (4\ 5) \\ (12) \quad \begin{pmatrix} 0.0 & & \\ 5.0 & 0.0 & \\ 8.0 & 4.0 & 0.0 \end{pmatrix} \\ 3 \\ (4\ 5) \end{array} \rightarrow \begin{array}{c} (12) \quad (3\ 4\ 5) \\ (12) \quad \begin{pmatrix} 0.0 & & \\ 5.0 & 0.0 & \end{pmatrix} \\ (3\ 4\ 5) \end{array}$$



A concrete example (min distance)

Finally, all data objects are grouped to one cluster.

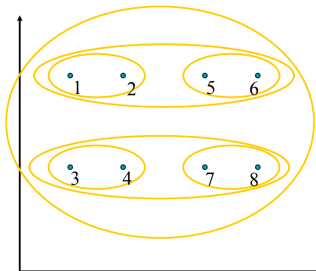
$$\begin{array}{c} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{array} \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 0.0 & & & & \\ 2.0 & 0.0 & & & \\ 6.0 & 5.0 & 0.0 & & \\ 10.0 & 9.0 & 4.0 & 0.0 & \\ 9.0 & 8.0 & 5.0 & 3.0 & 0.0 \end{pmatrix} \longrightarrow \begin{array}{c} (12) \quad (3 \ 4 \ 5) \\ (3 \ 4 \ 5) \end{array} \begin{pmatrix} 0.0 & \\ 5.0 & 0.0 \end{pmatrix}$$



Effect of cluster distance selection

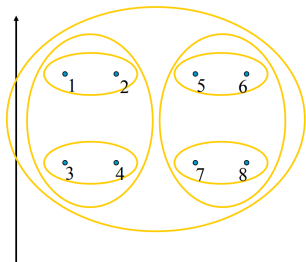
Closest pair

(single-link clustering)



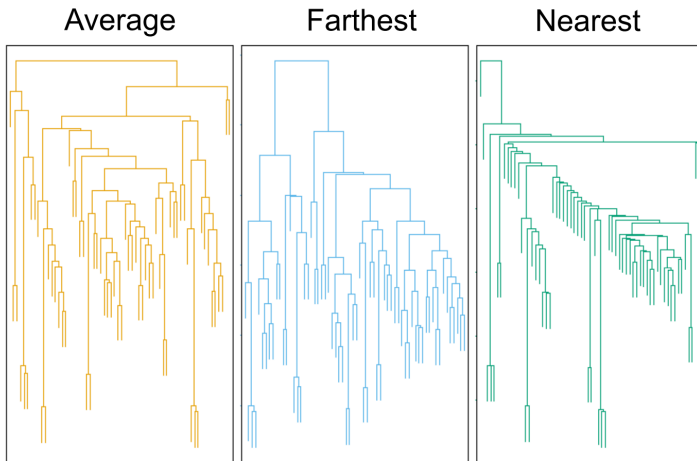
Farthest pair

(complete-link clustering)



[Pictures from Thorsten Joachims]

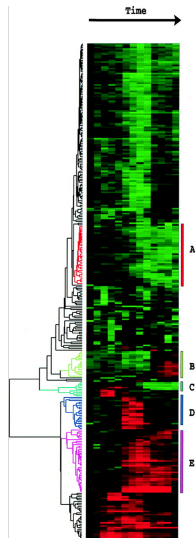
Effect of cluster distance selection



Mouse tumor data from [Hastie *et al.*]

Application example: clustering gene expression data

- Microarrays measures the activities of all genes in different conditions
- Clustering genes can help determine new functions for unknown genes



Eisen et al. PNAS 1998

Hierarchical clustering advantages

- Hierarchical clustering outputs a hierarchy, i.e., a structure that is more informative than the unstructured partition of flat clusters (e.g., that returned by K-Means or DBSCAN).
- Useful visualization tool, i.e., the dendrogram produced is useful for understanding the data.
- No need to specify the number of clusters required for the algorithm.
- Reproducible results, i.e., given the distant matrix and linkage criteria, the result will be same.
- Easy to implement.

Hierarchical clustering quiz

Suppose Cluster 1 contain $\left\{ \begin{bmatrix} -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\}$. Cluster 2 contain $\left\{ \begin{bmatrix} 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 5 \\ 0 \end{bmatrix} \right\}$. If we apply the single-linkage clustering, what is the distance between the two clusters.

- A. 0
- B. 1
- C. 2
- D. 3

Hierarchical clustering quiz

Suppose Cluster 1 contain $\left\{ \begin{bmatrix} -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\}$. Cluster 2 contain $\left\{ \begin{bmatrix} 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 5 \\ 0 \end{bmatrix} \right\}$. If we apply the single-linkage clustering, what is the distance between the two clusters.

- A. 0
- B. 1
- C. 2
- D. 3

Answer: B.

Reason: The minimum distance between the points in Cluster 1 and Cluster 2 is 1.

Hierarchical clustering disadvantages

- Cannot undo any previous steps, i.e., for example, the algorithm clusters two data points, but later we see that the connection is not a good one. In such situation, the program cannot undo that step.
- High time complexity, i.e., the following is a summary of hierarchical clustering complexity.

Method	Linkage criteria	Time complexity	Comment
Single Link	Smallest distance between two points	$O(n^2)$	Chaining effect
Complete Link	Longest Distance between two points	$O(n^2 \log n)$	Sensitive to noise
Average Link	Average Distance of all the data points	$O(n^2 \log n)$	Best choice for most applications
Centroid	Distance between Centroid	$O(n^2 \log n)$	Inversion can occur



Part 5. Summary

Summary

- Clustering has a long history and still is an active research area in unsupervised learning.
- Cluster analysis groups data points based on their similarity.
- Clustering is hard to evaluate, and highly application dependent.
- Understand popular algorithms
 - K-Means
 - DBSCAN
 - Agglomerative hierarchical clustering

Reading materials

- [Slides](#) on clustering by Carnegie Mellon University.
- A one-stop [website](#) on Data Clustering Algorithms.
- [Chapter 14.3](#), Cluster Analysis, The Elements of Statistical Learning (2nd Edition), Trevor Hastie, Robert Tibshirani, Jerome Friedman.